THE CATEGORY OF GRADED OBJECTS OF AN ABELIAN CATEGORY IS ABELIAN

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; both the existence of kernels and cokernels and the last axiom are written out, and the closing sentence is a conclusion rather than a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The same-name check for this card scanned every staged Stacks chapter of this repository – all sources/stacks-*/pinned-original-source.tex files, the card's own chapter excluded – and the labels found are recorded in dedup.same_name_cross_chapter on the card.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- Graded objects of an abelian category. Same chapter, the definition of the category of graded objects and of a morphism between them. Supplies the objects as families of objects indexed by degrees and the componentwise description of a morphism that the proof uses.
- Coimage and image in an abelian category. Same chapter, the coincidence of coimage and image for a morphism of an abelian category. Supplies the identity that is transported degree by degree to the category of graded objects.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. The category of graded objects $Gr(\mathcal{A})$ is abelian.*

Proof. Let $f : A = (A^i) \to B = (B^i)$ be a morphism of graded objects of $\mathcal{A}$ given by morphisms $f^i : A^i \to B^i$ of $\mathcal{A}$. Then we have $\mathrm{Ker}(f) = (\mathrm{Ker}(f^i))$ and $\mathrm{Coker}(f) = (\mathrm{Coker}(f^i))$ in the category $\mathrm{Gr}(\mathcal{A})$. Since we have $\mathrm{Im} = \mathrm{Coim}$ in $\mathcal{A}$ we see the same thing holds in $\mathrm{Gr}(\mathcal{A})$. □

Date: stacks-project:homology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.